

Program: Diploma in Wood and Paper Technology	
Course Code: 6121C	Course Title: Timber Structures
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4,T:0,P:0)	Periods per semester: 60

Course Objectives:

- To familiarize the structure of solid wood, timber and designing of beams.
- To provide knowledge on design of columns, joints, trusses more efficiently

Course Pre Requisites:

Topic	Course code	Course Title	Semester
Anatomy of wood		Botany and anatomical structures	2
Forest resources		Environmental Science	3

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain the structure of Solid Wood and Timber.	15	Understanding
CO2	Discuss the design Structure of Solid Timber and design of beam	15	Applying
CO3	Explain the design of Columns and Joints	15	Understanding
CO4	Summarize the design of Trusses and general design data	13	Applying
	Series Test	2	

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			2			
CO2			3		2		
CO3		3	2				
CO4		3					

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the structure of Solid Wood and Timber.		
M1.01	Structures of solid wood.	5	Remembering
M1.02	Understand soft wood and hard wood.	5	Remembering
M1.03	Structures of Timber.	5	Understanding
Contents: Structure of wood, comparison of softwood and hardwood, comparison of sapwood and heartwood, factors affecting strength of wood. - species, defects, moisture content, temperature, variability with in a species, duration of loading. Structure of timber, names of important Indian timbers,- density, shrinkage, tendency to warp, strength ,durability, availability &cost properties of timber, selection of timber- species, strength properties, physical properties, durability, availability			
CO2	Discuss the design Structure of Solid Timber and design of Beam		
M2.01	Design structures of solid timber.	3	Applying
M2.02	Extreme fiber stress in bending.	2	Understanding
M2.03	Basic working stresses.	2	Remembering
M2.04	Design stresses.	3	Applying
M2.05	Solid beams, Notched beams, Built up beams.	3	Remembering
M2.06	Timber-concrete beam sand slabs	2	Understanding
	Series Test-I	1	

Contents:

Design structure of solid timber, extreme fiber stress in bending, modulus of elasticity, compression parallel to grain, compression perpendicular to grain. Basic working stresses - grade and moisture content, design stresses, duration of loading, condition of exposure, angle of grain, width of bearing, rate of growth, diagram of typical nailed truss. Design of beams:- solid beam - fiber stress, form factor, longitudinal shear stress, stiffness, later stability, bearing stress (general idea only), solid beam, notched beams, built up beams, composite timber - concrete beams and slabs, design of laminated beams.

CO3	Explain the design of Column and Joints		
M3.01	Understand Design of Column.	4	Remembering
M3.02	Round columns, Spaced columns, Laminated columns.	4	Understanding
M3.03	Shrinkage effects with green timber and timber quality.	4	Remembering
M3.04	Factors affecting strength of glued joints.	3	Understanding

Contents:

Design of columns, types of columns- rectangular columns, effective column length, round columns, spaced columns, design of laminated columns. Design of joints, types of connections, metal connectors nail, screw and bolts, timber connectors, gluing. Shrinkage effects with green timber. Nailed joints, direction of loading, nails in end grain, moisture content, duration of loading. Multiple nail joints, nail penetration, splitting of timber, spacing of nails, allowable, withdrawal of loads. Bolted joints, multiple bolted joints, basic working loads parallel to grain and perpendicular to grain. Coach screws. Glued joints, factors affecting the strength of joints-timber quality, surface preparation, moisture content of wood and glue. Names of glues. Types of glued joints- butt joint, scarf joint, end joint, durability of glued joints

CO4	Summarize the design of trusses and general design data		
M4.01	Truss form and spacing	4	Understanding
M4.02	Stiffness of timber trusses.	4	Understanding
M4.03	Basic working stresses for glued laminated timber.	5	Applying
	Series Test II	1	

Contents:

Truss form and spacing, arrangement of members, number of joints, spacing of joints, spacing of trusses, stiffness of timber trusses, depth to span ratio, calculation of deflection. Basic working stresses for glued laminated timber (table). Design stresses, duration of load, number of laminations, moisture content, condition of exposure, angle of grain. General design data- end joints- scarf joint, butt joint, edge joint. Variation of grade within a member, mixed species, thickness of lamination, glues, direction of sawing laminae. Preservation of timber structure- general idea, preservation methods, types of preservatives. Economy of preservative treatment.

Text/Reference:

T/R	Book Title/Author
T1	R.G.PERSON, Timber Engineering design handbook, Thejacaranda press Australia
T2	F.P.KOLLMAN, Principles of wood science technology volume I&II, Springerverlag

Online Resources:

Sl. No	Website Link
1	https://www.archdaily.com/tag/timber-construction
2	https://www.woodsolutions.com.au/blog/worlds-largest-timber-structures
3	https://www.youtube.com/watch?v=LWGrLZ_Js2U
4	https://www.bfh.ch/en/research/research-areas/institute-timber-construction-structures-architecture-ihta/